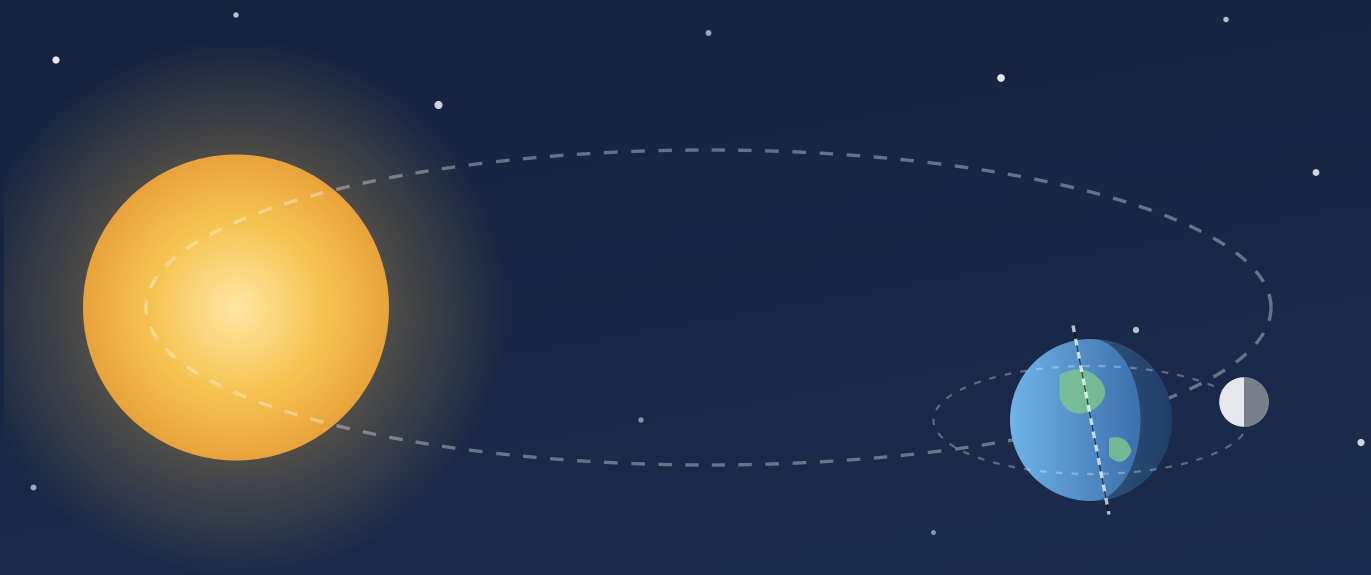
GRADE 4 SCIENCE · LESSON BOOK

UNIT 2

Earth, Moon and Sun

Why the sky changes: day and night, seasons, moon phases and tides



11 lessons
Lessons 15 to 25, two pages each

1 spiral check
ungraded, after Lesson 18

1 unit review
with a CER writing task

Full answer key
for honest self-checking

Why is it night on the other side of the world?

Why are there seasons?

Why does the moon change shape?

Why are some tides bigger?

Name _____ Class _____ Teacher _____

Aligned to the Virginia Science Standards of Learning 4.1 and 4.6

How this lesson book is designed

Every lesson in this book is built to be taught by any teacher and completed by any student, including a student working alone. Five design decisions make that possible.

1 One routine, eleven times

Every lesson is exactly two pages and every page does the same jobs in the same order. Students stop spending effort on how to use the page and spend it on the science. The routine is identical to Unit 1.

2 Read first, then do

Page 1 teaches the idea through a short passage split into numbered parts, with vocabulary given before the reading. Every Check Your Reading question names the part its answer sits in, so a wrong answer sends the student back to a specific paragraph rather than to a guess.

3 Gradual release on page 2

A worked example shows the thinking for one item, the student applies the same method to the rest, and a hint is held back until after an honest attempt. The method is modeled; the answer is never given.

4 Misconceptions named out loud

The errors this unit is known for (seasons caused by distance, phases caused by Earth's shadow, day and night caused by revolution) are each raised in the reading, tested by a question and flagged in a Careful box. Lesson 21 asks students to find and correct a classmate's mistakes.

5 Retrieval and self-assessment

An ungraded Spiral Check after Lesson 18 revisits the first four lessons before the moon is introduced. Every lesson closes with a three-item checklist, and the answer key at the back comes with three rules for honest self-marking.

6 Built for every learner

Every lesson carries a hint for students who need support and most carry a challenge for students ready for more. Diagrams carry the idea without relying on reading alone, and nothing in the book needs equipment beyond a pencil.

STANDARDS CORRELATION

Standard	What students do in this unit	Lessons
VA SOL 4.6a motions of Earth, the moon and the sun	Explain day and night through rotation; follow one place through a full rotation; explain time zones	15, 16, 24
VA SOL 4.6b causes for Earth's seasons	Explain seasons through tilt plus revolution; use the Southern Hemisphere to rule out distance; compare direct and indirect rays	17, 18, 24
VA SOL 4.6c four major phases of the moon and the tide cycle	Name and order the four phases; match phase to orbital position; explain two tides a day and the monthly change in tidal range	19 to 23
VA SOL 4.1 scientific and engineering practices	Use models; analyze and interpret data; construct and critique explanations; keep a consistent record over time	16, 21, 23, 24, 25
VA SOL 4.4 weather (prepares Unit 3)	Begin a nine-week weather log using a consistent method	25

Also supports NGSS 5-ESS1-2 (patterns of change in the Earth, sun and moon system).

How every lesson works

Every lesson takes two pages, and the two pages always do the same jobs. Learn the pattern once and you can use it all year.

PAGE 1 · Learn the idea

By the end of this lesson I can
Read this first. It tells you what you are aiming at.

Words used in this lesson
Read the meanings before the passage, not after.

Read this first
The passage, split into numbered parts. Read it once, then again with a pencil, underlining one sentence in each part.

Check your reading The number after each question names the part the answer is in. If you cannot find it, go back to that part. Do not guess.

PAGE 2 · Use the idea

Worked example
One question already done, with the thinking shown. Copy the method, not the answer.

Now you try
The activity, broken into numbered steps. Do them in order.

If you get stuck
A hint. Use it after you try.

Challenge
For when you finish early.

Before you finish
Three things to check. If you cannot check one honestly, go back.

WORKING ON YOUR OWN

Nothing in this book needs a teacher standing next to you, and nothing needs equipment you do not have. Every picture, every number and every instruction is printed here.

Draw all models in pencil. Scientists change their models when new evidence arrives, and so will you.

Keep Unit 1 in your folder. This unit uses it all the way through, especially rotation and revolution from Lesson 10. That is why the lessons here are numbered 15 to 25.

CHECKING YOUR OWN WORK HONESTLY

1. **Write first, check second.** A blank you fill in from the key teaches you nothing.
2. **Same idea, different words is right.** Science cares about the idea, not the exact wording.
3. **Do not erase a wrong answer.** Cross it out, write the correct one beside it, and add a short note saying why yours was wrong. The answers you fix yourself are the ones you remember.

BY THE END OF THIS UNIT I WILL BE ABLE TO

- Explain why we have day and night, and why it is not the same time everywhere on Earth.
- Explain the seasons using tilt and revolution together, and show why distance from the Sun is not the cause.
- Explain why light arriving at a slant heats a surface less than light arriving straight down.
- Name the four moon phases in order and match each one to the moon's position on its orbit.
- Explain why there are two high tides in about a day, and when the tidal range is largest and smallest.
- Calculate tidal ranges from a table, find a pattern in real data, and use it to predict.

Unit 2 at a glance

The unit runs across sixteen class days. Eleven are lessons in this book. The other five are a spiral check, a catch-up day, a review day, the unit test and one day after the test.

Day	Lesson	Title	What happens
1	Lesson 15	Day and night	Rotation, sunrise and sunset, why clocks differ around the world
2	Lesson 16	Where am I when it is night?	Follow one place through a full rotation; why time zones exist
3	Lesson 17	Why do we have seasons?	Tilt plus revolution; Australia rules out distance
4	Lesson 18	Direct and indirect rays	Same light, bigger area, less heating
5	Spiral Check	Lessons 15 to 18	Six questions, ungraded, checked in class
6	Lesson 19	The four moon phases	Name and order the phases; the moon never changes shape
7	Lesson 20	Moon positions and phases	The lit half always faces the Sun
8	Lesson 21	Fix the diagram	Why the shadow idea fails; correct a classmate's work
9	Lesson 22	Tides	Two bulges, two high tides; alignment and tidal range
10	Lesson 23	Reading tidal data	Calculate ranges, find the pattern, predict
11	Lesson 24	How it all connects	Cause and effect concept map of the whole system
12	Lesson 25	Weekly weather log	Start a nine-week record that carries into Unit 3
13	Catch-up	No new content	Finish unfinished lessons; redo Check Your Reading
14	Unit Review	Show what you know	Review pages worked through in class, including the CER task
15	Unit Test	30 points, full period	Summative assessment
16	After the test	Test returned	Correct errors with notes; weather log Week 2

IF THE CLASS FALLS BEHIND

The two days to give up are Day 5 (the spiral check, which can move into a warm-up) and Day 13 (catch-up). Never give up the review day and never run the test in half a period.

THE ONE LESSON THAT CANNOT MOVE

Lesson 25 starts a nine-week weather log that feeds Unit 3. If everything else slips, that lesson still happens on Day 12, because the log needs the weeks.

LEARN THE IDEA

15 Day and night

Name _____

Date _____

BY THE END OF THIS LESSON I CAN

1. Explain why we have day and night, using the word **rotation**.
2. Explain why it is not the same time everywhere on Earth.

WORDS USED IN THIS LESSON

rotation
Earth spinning on its axis, once in about 24 hours

sunrise
the moment your place turns into the sunlight

sunset
the moment your place turns out of the sunlight

time zone
an area of Earth that keeps the same clock time

READ Why the Sun seems to move

Read it all once. Then read it again and underline one sentence in each part.

Part 1

We are the ones moving

It looks as though the Sun crosses the sky from one side to the other. It does not. We are the ones moving.

Earth spins on its axis, making one full turn in about 24 hours. At any moment, half of Earth faces the Sun and has daylight, while the other half faces away and has night. As Earth keeps turning, your part of the world is carried from the lit side into the dark side and back again.

Part 2

Sunrise and sunset

Sunrise is the moment your place turns into the light. Sunset is the moment it turns out of it. Nothing about the Sun has changed during either event.

This is worth saying carefully, because everyday words work against you. We say the Sun rises and the Sun sets. Those words describe what it looks like, not what happens.

Part 3

Why the time is different elsewhere

This is also why it is not the same time everywhere on Earth. When it is morning in Virginia, it is already evening in Japan, because the two places are at different points in the same rotation. Every place gets its turn in the light.

Day and night come from Earth spinning, not from Earth traveling around the Sun. Mixing those two up is the most common error in this unit.

CHECK YOUR READING Every answer is in the passage. The part number tells you where to look.

1. What is really moving when the Sun appears to cross the sky? (Part 1)

2. How much of Earth is in daylight at any moment? (Part 1)

3. What is happening at the moment we call sunset? (Part 2)

4. Why is it not the same time of day everywhere on Earth? (Part 3)

USE THE IDEA

15 Day and night

WORKED EXAMPLE Study this before you write anything. Copy the method, not the answer.

Question: Which side of a day and night diagram is having day?

Step 1. Find where the sunlight is coming from. In this example the arrows come from the left.

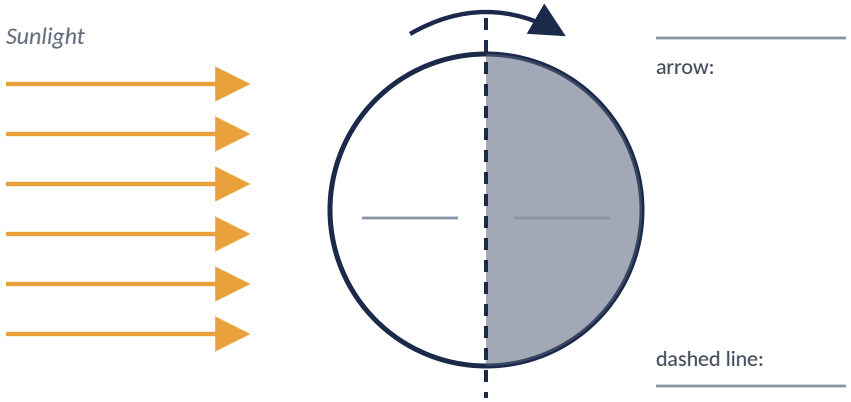
Step 2. The half of Earth facing those arrows is lit. That is the left half.

Step 3. So the left half is day and the right half is night. Label them that way.

Use the same three steps on any day and night diagram, whichever side the light comes from.

NOW YOU TRY

STEP 1 Label the diagram. Write DAY and NIGHT on the two lines inside Earth. Then name the dashed line and the curved arrow on the lines beside them.



STEP 2 Why do we have day and night? Answer in one full sentence that uses the word **rotation**.

STEP 3 It is 7 in the morning where you are. Is it 7 in the morning everywhere on Earth? Explain why.

IF YOU GET STUCK

Cover the diagram and say this out loud: the Sun lights the half of Earth that faces it. Now uncover it. Which side is facing the arrows? That side is day.

CAREFUL: A COMMON MISTAKE

Day and night come from ROTATION, not revolution. Revolution gives us the year. If your Step 2 sentence says Earth goes around the Sun, it is wrong.

BEFORE YOU FINISH, CHECK EACH ONE

- [ ] I labeled day, night, the axis and the direction of rotation.
- [ ] My Step 2 sentence contains the word rotation or spinning.
- [ ] My Step 3 answer explains that different places face the Sun at different moments.

LEARN THE IDEA

16 Where am I when it is night?

Name _____

Date _____

BY THE END OF THIS LESSON I CAN

1. Follow one place through a whole rotation and say the time of day at each position.
2. Explain why the world needs time zones.

WORDS USED IN THIS LESSON

position — where something is at one particular moment

midnight — when your place is turned farthest from the sunlight

midday — when your place faces the Sun most directly

time zone — an area of Earth that keeps the same clock time

READ Following one place around

Read it all once. Then read it again and underline one sentence in each part.

Part 1 **Watch one place, not the whole globe**

A globe is much easier to understand if you stop looking at the whole thing and follow one single place instead.

Put a sticker on Virginia. Shine a lamp from one side and turn the globe slowly. Watch only the sticker. It moves into the light, travels across the lit side, slips into the dark, travels across the night side, and comes back to the light again.

Part 2 **One sticker, one whole day**

That one sticker is having a whole day. When it first reaches the light, it is sunrise. When it faces the lamp most directly, it is about midday. When it leaves the light, it is sunset. When it is turned farthest from the light, it is the middle of the night.

The lamp does not move and the globe does not travel anywhere, and still the sticker gets a morning, an afternoon and a night.

Part 3 **Why time zones exist**

Every place on Earth is doing this at the same time, but each place reaches the light at a different moment. A place to the east of you turns into the light before you do. A place to the west turns into it after you.

That is exactly why the world needs time zones. If every place kept the same clock, midday would fall in the middle of the night for somebody.

CHECK YOUR READING Every answer is in the passage. The part number tells you where to look.

1. When is it midday for the sticker? (Part 2)

2. What is happening to the sticker at the moment we call sunrise? (Part 2)

3. Why does the world need time zones? (Part 3)

4. Does the lamp have to move for the sticker to have a whole day? Explain. (Part 2)

USE THE IDEA

16 Where am I when it is night?

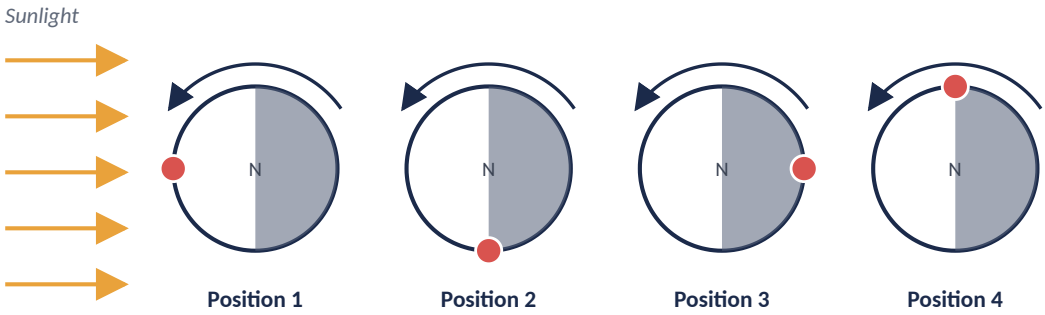
WORKED EXAMPLE Study this before you write anything. Copy the method, not the answer.

These globes are drawn **looking down from above the North Pole**. The red dot is your town. The curved arrow shows which way Earth turns.

Position 1: Is the dot in the lit half or the dark half? *The lit half.* What time of day is it? *About midday.* How do I know? *The dot faces the sunlight directly, and that is what midday means.*

Work the other three the same way: lit or dark first, then the time, then the reason.

NOW YOU TRY



STEP 1 Fill in the three empty rows. Use the turning arrow to decide which comes first after midday.

Position	Lit half or dark half?	What time of day?	How I know
1	lit	about midday	the dot faces the sunlight directly
2			
3			
4			

IF YOU GET STUCK

Position 3 is the opposite of Position 1. What is the opposite of midday? Positions 2 and 4 are in between, so one is sunset and one is sunrise. Follow the arrow from Position 1 to see which the dot reaches first.

CHALLENGE

A friend in another country calls you at 8 in the morning your time and says it is already evening where they are. Are they east or west of you? Explain how you decided.

BEFORE YOU FINISH, CHECK EACH ONE

- ☐ I filled in all three empty rows of the table.
- ☐ Every reason in the last column says where the dot is compared with the sunlight.
- ☐ I used the turning arrow to decide the order of sunset and sunrise.

LEARN THE IDEA

17 Why do we have seasons?

Name _____

Date _____

BY THE END OF THIS LESSON I CAN

1. Explain that seasons are caused by Earth's tilt together with its revolution.
2. Use Australia to show that distance from the Sun cannot be the cause.

WORDS USED IN THIS LESSON

tilt
the lean of Earth's axis, which points the same way all year

revolution
Earth's yearly trip around the Sun

hemisphere
one half of Earth, the northern half or the southern half

toward / away
leaning in the direction of something, or the opposite direction

READ It is the tilt, not the distance

Read it all once. Then read it again and underline one sentence in each part.

Part 1

The idea that sounds right but is not

Many people believe summer happens because Earth is closer to the Sun. That is not the reason, and there is an easy way to see why.

When it is summer in Virginia, it is winter in Australia. If distance from the Sun were the cause, the whole planet would have the same season at the same time, because the whole planet is the same distance away. It does not, so distance cannot be the cause.

Part 2

The real cause

The real cause is that Earth is tilted. Its axis leans over, and it keeps leaning in the same direction all year long while Earth travels around the Sun.

For part of the year the northern half of Earth leans toward the Sun. That half receives more direct sunlight and has longer days, so it warms up. That is summer. Six months later Earth is on the other side of its orbit. The same northern half now leans away, sunlight arrives at a lower angle, days are shorter, and it is winter.

Part 3

What actually changes

The tilt never changes direction during the year. What changes is where Earth is on its journey, and so which half happens to be leaning toward the light.

That is why seasons need two things together: the tilt and the revolution. Neither one on its own would produce a season.

CHECK YOUR READING Every answer is in the passage. The part number tells you where to look.

1. What season is it in Australia when it is summer in Virginia? (Part 1)

2. Use that fact to explain why distance from the Sun cannot cause the seasons. (Part 1)

3. What two things together cause the seasons? (Part 3)

4. Does the direction of Earth's tilt change during the year? What changes instead? (Part 3)

USE THE IDEA

17 Why do we have seasons?

WORKED EXAMPLE Study this before you write anything. Copy the method, not the answer.

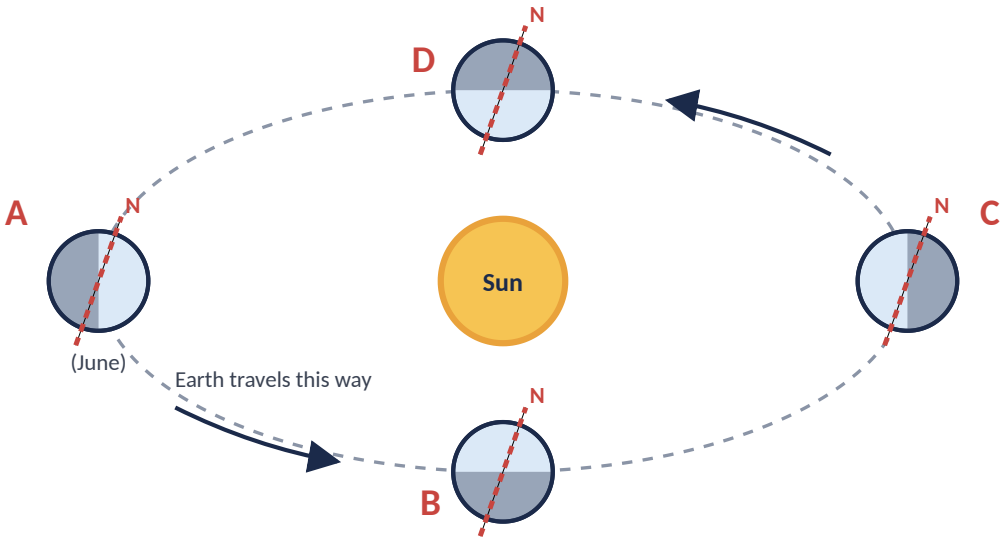
How to read the tilt at any position:

Step 1. Find the red dashed axis line through Earth. N marks the top, the North Pole.

Step 2. Look at which way the top of that line leans.

Step 3. If the top leans toward the Sun, the northern half is having summer. If it leans away, the northern half is having winter. If it leans neither toward nor away, it is spring or fall. Follow the travel arrow: the season after summer is fall.

NOW YOU TRY



STEP 1 Label each position: **Northern summer, Fall, Northern winter, Spring.**

A _____ B _____ C _____ D _____

STEP 2 Position A is June. What season is it in Australia in June? Explain how you know.

IF YOU GET STUCK

Look at the top of each dashed line. Toward the Sun means northern summer. Away means northern winter. Start at summer and follow the arrow: summer, then fall, then winter, then spring.

CAREFUL: A COMMON MISTAKE

Seasons are caused by TILT plus REVOLUTION, not by distance from the Sun. This is the most common wrong answer in Grade 4 science, so check your sentences for the word *closer* before you finish.

BEFORE YOU FINISH, CHECK EACH ONE

- ☐ I labeled all four positions on the diagram.
- ☐ My answer for Australia is the opposite season from Virginia.
- ☐ None of my answers says summer happens because Earth is closer to the Sun.

LEARN THE IDEA

18 Direct and indirect rays

Name _____

Date _____

BY THE END OF THIS LESSON I CAN

1. Explain why light striking at a slant heats a surface less than light striking straight down.
2. Connect the angle of the rays to the season we are having.

WORDS USED IN THIS LESSON

direct rays — light arriving straight down, packed onto a small area

indirect rays — light arriving at a slant, spread thin over a large area

concentrated — packed into a small space

spread — shared out over a larger space, so each part gets less

READ Why the angle of the light matters

Read it all once. Then read it again and underline one sentence in each part.

Part 1 **The flashlight on the table**

Hold a flashlight straight down over a table and you get a small, bright circle of light. Tilt the flashlight over and the same light spreads out into a wide, dim oval.

The flashlight is not making more or less light. The same amount of light is being spread over a different amount of surface. When it is concentrated on a small area, that area heats up more. When it is spread thin, each part of the surface receives less.

Part 2 **Sunlight works the same way**

The half of Earth tilted toward the Sun receives direct rays, concentrated on a smaller area, so it warms more. The half tilted away receives indirect rays, spread thin over a larger area, so it warms less.

Nothing about the Sun changes between summer and winter. The Sun does not get hotter in June. Only the angle at which its light arrives changes.

Part 3 **The second effect**

A second effect works alongside the angle. The half tilted toward the Sun also has longer days, so it has more time to warm up and less time to cool down at night.

Direct rays and long days together are what make summer warm.

CHECK YOUR READING Every answer is in the passage. The part number tells you where to look.

1. When you tilt a flashlight, does the light get weaker, or does something else change? (Part 1)

2. Which rays heat a surface more, direct or indirect? Why? (Part 1)

3. Does the Sun get hotter in summer? What changes instead? (Part 2)

4. Besides the angle of the rays, what else makes summer warm? (Part 3)

USE THE IDEA

18 Direct and indirect rays

WORKED EXAMPLE Study this before you write anything. Copy the method, not the answer.

The same flashlight was used for both grids, so exactly the same amount of light landed on each.

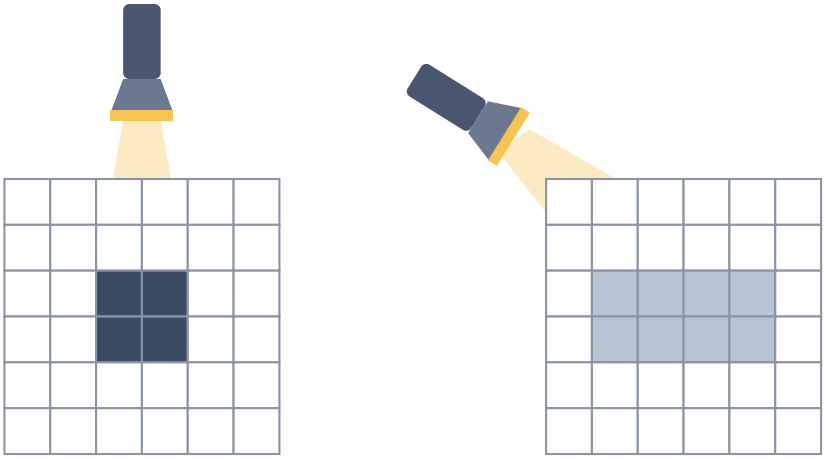
Grid A: the light falls on a few squares, and those squares are shaded dark.

Grid B: the same light falls on more squares, and every square is shaded lighter.

Reasoning: if the same light is spread over more squares, each square must be getting less. That is why B is lighter, and it is why indirect rays heat a surface less.

NOW YOU TRY

A: light straight down **B: light at a slant**



The same flashlight was used both times.

STEP 1 Count the lit squares in each grid and fill in the table.

	A: straight down	B: at a slant
How many squares are lit? (count them)		
Is each lit square dark gray or light gray?		
Which grid's squares each get more light?		
Which grid heats up more?		
Which one is like summer where we live?		

STEP 2 The hemisphere tilted TOWARD the Sun gets _____ rays and has _____ days.

IF YOU GET STUCK

The same amount of light landed on both grids. If grid B spreads that light over MORE squares, each square must be getting LESS.

CHALLENGE

Why is it usually warmer at midday than at breakfast time, even on the same day? Use the words direct and indirect in your answer.

BEFORE YOU FINISH, CHECK EACH ONE

- ☐ I counted the squares in both grids rather than guessing.
- ☐ My table says grid A heats up more.
- ☐ My Step 2 sentence uses the word direct.

SPIRAL CHECK · LESSONS 15 TO 18

What do I remember?

Name _____

Date _____

This check is not graded. It shows you and your teacher what has stuck before the moon arrives in Lesson 19. Answer from memory first. Then, for anything you are unsure of, find the lesson and page that would help and write its number in the last column.

1. Which motion of Earth causes day and night? Circle one.

   A. revolution B. rotation C. tilt D. the Sun moving across the sky

   Lesson I would check: ____

2. It is noon in Virginia. Is it noon in Japan too? Explain in one sentence.

   _____

   _____

   Lesson I would check: ____

3. A classmate says, "It is hot in summer because Earth is closer to the Sun." Write one sentence that proves this wrong.

   _____

   _____

   Lesson I would check: ____

4. Which two things **together** cause the seasons? Circle two.

   A. tilt B. distance C. revolution D. rotation

   Lesson I would check: ____

5. Complete the sentence: Direct rays heat a surface more than indirect rays because the same light is

   _____

   _____

   Lesson I would check: ____

6. A town is on the side of Earth turned farthest from the Sun. About what time is it there? How do you know?

   _____

   _____

   Lesson I would check: ____

AFTER YOU CHECK YOUR ANSWERS

- ☐ I circled the number of every question I got wrong.
- ☐ For each one, I reread the part of the lesson I named and fixed my answer with a note.

19

LEARN THE IDEA

The four moon phases

Name _____

Date _____

BY THE END OF THIS LESSON I CAN

1. Name the four main moon phases and put them in order.
2. Explain that the moon does not change shape, only how much of its lit half we can see.

WORDS USED IN THIS LESSON

phase — how much of the moon's lit half we can see from Earth

full moon — the phase when we see all of the lit half

new moon — the phase when we cannot see the lit half at all

cycle — a set of changes that repeats in the same order

READ The moon does not change shape

Read it all once. Then read it again and underline one sentence in each part.

Part 1 What is really changing

The moon looks different from one night to the next, but the moon itself never changes. It is always a ball, and the Sun always lights exactly half of it.

What changes is how much of that lit half we can see from Earth, and that depends on where the moon is on its trip around us. The moon is not growing and shrinking. We are looking at it from a different angle each night.

Part 2 The four phases you need

There are four main phases to know. At new moon we cannot see the lit half at all, so the moon is almost invisible. At first quarter we see half of the lit side. At full moon we see all of it. At last quarter we see the other half.

In order they run: new moon, first quarter, full moon, last quarter, and then back to new moon. It is a cycle, so it has no real beginning, but we usually start counting at new moon.

Part 3 How long, and the far side

The moon takes about 29 days to go through all its phases, so the whole cycle takes about a month. In fact, the word month comes from the word moon.

We always see the same side of the moon from Earth. The far side is not dark. It simply never faces us, and nobody had seen it until spacecraft flew around the back of it.

CHECK YOUR READING Every answer is in the passage. The part number tells you where to look.

1. Does the moon really change shape? What is actually changing? (Part 1)

2. Write the four phases in the order they happen. (Part 2)

3. About how many days does the whole cycle of phases take? (Part 3)

4. Why had nobody seen the far side of the moon until spacecraft went there? (Part 3)

USE THE IDEA

19 The four moon phases

WORKED EXAMPLE Study this before you write anything. Copy the method, not the answer.

Look at picture 1. It is completely dark, with none of the lit half showing.

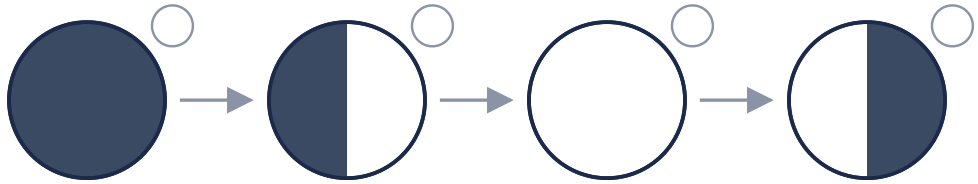
That means we cannot see any of the sunlit side from Earth, so it is a **new moon**, and it is number 1 in the order.

Always work out what is lit and what is dark first, then name the phase. Never name the phase first and then make the picture fit.

NOW YOU TRY

STEP 1 Write the name of each phase on the line underneath it.

STEP 2 Number them 1 to 4 in the small circles, in the order they happen.



_____ _____ _____ _____

Word bank: new moon · first quarter · full moon · last quarter

STEP 3 Write each word in your own words.

Word	What it means in my own words
phase	
new moon	
full moon	
cycle	

IF YOU GET STUCK

The four names are in Part 2 of the reading, already in the right order. The pictures are drawn in that same order, from left to right.

DID YOU KNOW?

Before clocks and printed calendars, counting moons was one of the ways people kept track of the passing year.

BEFORE YOU FINISH, CHECK EACH ONE

- ☐ I named all four phases on the diagram.
- ☐ I numbered them 1 to 4 in the correct order.
- ☐ I can explain that the moon is always a ball with half of it lit.

LEARN THE IDEA

20 Moon positions and phases

Name _____

Date _____

BY THE END OF THIS LESSON I CAN

1. Match each position of the moon on its orbit to the phase we see from Earth.
2. Remember that the lit half of the moon always faces the Sun.

WORDS USED IN THIS LESSON

orbit — the moon's path around Earth

position — where the moon is on its orbit at one moment

lit half — the half of the moon the Sun shines on

viewing angle — the direction we look at the moon from

READ **Where the moon is decides what we see**

Read it all once. Then read it again and underline one sentence in each part.

Part 1 **The lit half always faces the Sun**

The lit half of the moon always faces the Sun. Always. It does not matter where the moon is on its orbit, and it does not matter what phase we happen to see from here.

This one sentence is the whole lesson. If you hold on to it, every question on this page can be worked out rather than remembered.

Part 2 **What changes is our viewing angle**

When the moon is between Earth and the Sun, its lit half points away from us and we see only the unlit side. That is a new moon. When the moon is on the far side of Earth from the Sun, the lit half faces us completely, and we see a full moon.

At the two positions in between, we see the lit half from the side, so half of what faces us is lit and half is dark. Those are the quarter phases. To know which quarter comes first, follow the arrow showing the way the moon travels.

Part 3 **The mistake to watch for**

The most common mistake is shading a drawn moon so that its lit half points toward Earth, because that is the half we are interested in.

Check every moon you draw: the bright side must point at the Sun, even when that means it points away from Earth. What Earth can see is the second question, never the first.

CHECK YOUR READING Every answer is in the passage. The part number tells you where to look.

1. Which way does the lit half of the moon face? Is that ever different? **(Part 1)**

2. Where is the moon when we see a new moon? **(Part 2)**

3. Where is the moon when we see a full moon? **(Part 2)**

4. What is the most common mistake students make on this diagram? **(Part 3)**

USE THE IDEA

20 Moon positions and phases

WORKED EXAMPLE Study this before you write anything. Copy the method, not the answer.

Position A: the moon sits between Earth and the Sun.

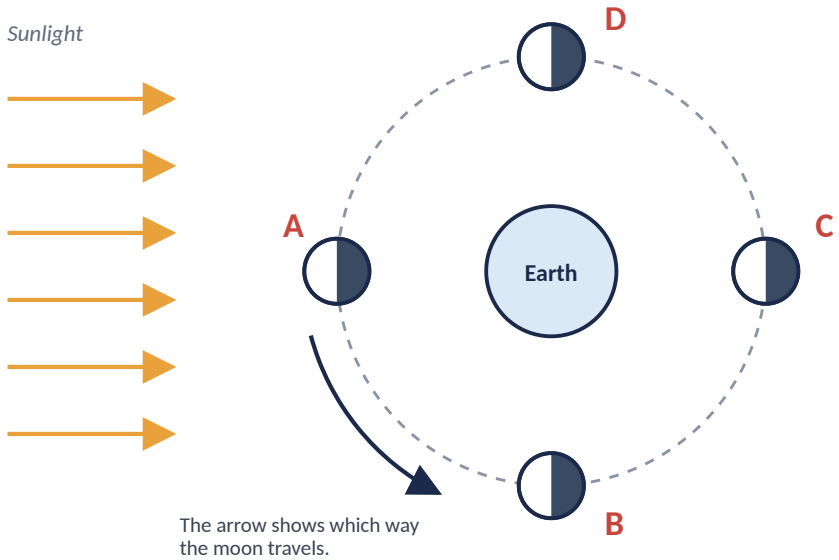
Where is its lit half pointing? *Left, toward the Sun, which is away from Earth.*

What can Earth see from there? *Only the unlit half.*

Which phase is that? *A new moon.*

Ask the same three questions for B, C and D, in that order, every time.

NOW YOU TRY



STEP 1 Fill in the phase for positions B, C and D.

STEP 2 In the last column, write how you worked it out. Say where the lit half was pointing.

Position	Phase we see from Earth	How I worked it out
A	new moon	lit half points away from Earth
B		
C		
D		

IF YOU GET STUCK

Position C is directly opposite the Sun, so Earth sees the whole lit half. Do that one first. B and D are the halfway positions. The arrow tells you which of them the moon reaches first after A.

CAREFUL: A COMMON MISTAKE

Shade the lit half toward the SUN, not toward Earth. What Earth can see is the second question, never the first.

BEFORE YOU FINISH, CHECK EACH ONE

- ☐ I filled in the phase for B, C and D.
- ☐ Every reason I wrote says where the lit half was pointing.
- ☐ I did position C first because it was the easiest.

LEARN THE IDEA

21 Fix the diagram

Name _____

Date _____

BY THE END OF THIS LESSON I CAN

1. Explain why the phases are not caused by Earth's shadow.
2. Find and correct mistakes in somebody else's answer.

WORDS USED IN THIS LESSON

shadow — the dark area behind something that is blocking light

viewing angle — the direction we look at something from

lunar eclipse — the rare event when Earth's shadow really does fall on the moon

correct — to find a mistake and write what it should have been

READ Phases are not a shadow

Read it all once. Then read it again and underline one sentence in each part.

Part 1

A very sensible wrong idea

A very sensible wrong idea is that the phases are Earth's shadow falling on the moon. It sounds right, and almost everybody thinks it at first.

It is worth taking seriously, because working out exactly why it is wrong teaches you more than simply being told the right answer.

Part 2

Why the shadow idea fails

Earth's shadow always points in one direction: away from the Sun. If that shadow caused the phases, the moon would look the same every night, because the shadow would always fall in the same place.

But the moon clearly changes over the month, so the shadow cannot be the cause. Notice how that argument works: take the idea, follow it through to what we would see, and compare that with what we actually see.

Part 3

When the shadow does fall on the moon

Earth's shadow does sometimes fall on the moon. That is called a lunar eclipse, and it is rare: a few times a year at most, and it is over within a few hours.

So the two things are different. Phases come from our viewing angle as the moon travels around us. Eclipses come from a shadow. If you can explain that difference clearly, you understand this topic better than most adults do.

CHECK YOUR READING Every answer is in the passage. The part number tells you where to look.

1. If Earth's shadow caused the phases, what would we see every night? (Part 2)

2. What is it called when Earth's shadow really falls on the moon, and how often does it happen? (Part 3)

3. In one sentence, what actually causes the phases? (Part 3)

4. Describe the way of testing an idea that Part 2 uses. (Part 2)

USE THE IDEA

21 Fix the diagram

WORKED EXAMPLE Study this before you write anything. Copy the method, not the answer.

How to find mistakes in somebody's order.

Correct order: new moon, first quarter, full moon, last quarter.

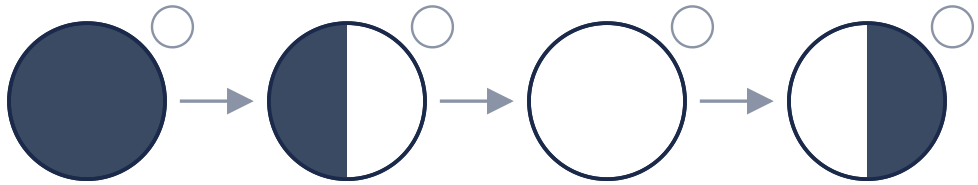
Student A wrote: *first quarter, new moon, full moon, last quarter.*

Compare one position at a time. Position 1 should be new moon, but Student A wrote first quarter. That is mistake 1. Position 2 should be first quarter, but Student A wrote new moon. That is mistake 2. Positions 3 and 4 match.

Always write the mistake AND what it should have been. "It is wrong" on its own earns no points.

NOW YOU TRY

Warm up: write the four phase names under the pictures again, in order. Then start Step 1.



Student B wrote the phases in this order: new moon, full moon, first quarter, last quarter.

STEP 1 Mistake 1, and what it should have been:

STEP 2 Mistake 2, and what it should have been:

STEP 3 Student B then wrote: *"The phases happen because Earth's shadow falls on the moon."* Write one sentence explaining why that is wrong.

IF YOU GET STUCK

Write the correct order first, from Lesson 19. Then compare it with Student B's order, one position at a time. The mistakes are wherever the two lists disagree.

CHALLENGE

If the moon takes about 29 days to go through all its phases, about how many days pass between a new moon and the next full moon? Show how you worked it out.

BEFORE YOU FINISH, CHECK EACH ONE

- [ ] For each mistake I wrote what was wrong AND what it should have been.
- [ ] I compared the two lists one position at a time.
- [ ] My Step 3 sentence explains why the shadow idea fails, not just that it is wrong.

LEARN THE IDEA

22 Tides

Name _____

Date _____

BY THE END OF THIS LESSON I CAN

1. Explain why most coasts get two high tides in about a day.
2. Say at which moon phases the tidal range is largest and smallest.

WORDS USED IN THIS LESSON

tide — the daily rise and fall of the sea along a coast

tidal range — the difference in height between high tide and low tide

bulge — a place where ocean water is pulled up higher

alignment — how the Sun, Earth and moon are lined up

READ Two bulges of water

Read it all once. Then read it again and underline one sentence in each part.

Part 1 Two high tides a day

The ocean does not sit still. Twice a day the water rises up the beach, and twice a day it falls back again. Those are the tides, and people who live and work by the sea plan their day around them.

The moon pulls on Earth's oceans. That pull draws the water into a bulge on the side of Earth facing the moon, and there is a second bulge on the opposite side. As Earth rotates, every coast passes through both bulges. That is why most places get two high tides and two low tides in about a day.

Part 2 Tidal range

The difference in height between high tide and low tide is called the tidal range. You find it by subtracting the low tide height from the high tide height.

The range is not the same every day, and that is the part worth explaining. Something must be changing from one week to the next, and it is not the ocean.

Part 3 Why the range changes

The Sun pulls on the oceans too, more weakly than the moon but enough to matter. What changes through the month is whether the two pulls line up.

When the Sun, Earth and moon are in a line, at new moon and full moon, the two pulls work together and the tidal range is at its largest. When the moon sits at a right angle to the Sun, at the two quarter phases, the pulls work partly against each other and the range is at its smallest.

CHECK YOUR READING Every answer is in the passage. The part number tells you where to look.

1. Why does one place get two high tides in about a day? (Part 1)

2. What is tidal range, and how do you work it out? (Part 2)

3. Why is the range largest at new moon and full moon? (Part 3)

4. At which two phases is the range smallest? (Part 3)

22 USE THE IDEA

Tides

WORKED EXAMPLE Study this before you write anything. Copy the method, not the answer.

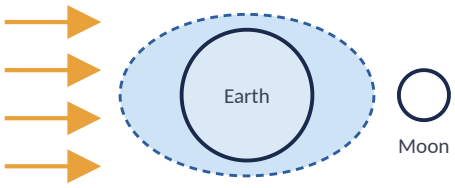
Look at the left drawing. The Sun, Earth and moon make a straight line.

Reasoning: lined up means the Sun's pull and the moon's pull work in the same direction, so together they pull harder than either could alone. The water bulges more, so the tidal range is LARGEST.

The right drawing makes an L shape instead. The two pulls work partly against each other, so the range is SMALLEST.

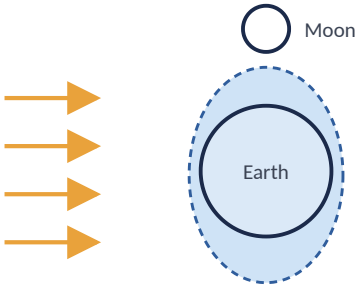
NOW YOU TRY

Sun, Earth and moon in a LINE
new moon and full moon



the two pulls work together:
LARGEST tidal range

Moon at a RIGHT ANGLE
first quarter and last quarter



pulls work partly against each other:
SMALLEST tidal range

STEP 1 The largest tidal range happens at which two phases?

STEP 2 The smallest tidal range happens at which two phases?

STEP 3 Write each word in your own words.

Word	What it means in my own words
tide	
tidal range	
alignment	

IF YOU GET STUCK
In the left drawing the three bodies make a straight line. In the right one they make an L. Part 3 of the reading tells you which of those gives the biggest range.

CAREFUL: A COMMON MISTAKE
There are two bulges, not one, which is why there are two high tides a day. The second bulge is on the side of Earth facing away from the moon.

BEFORE YOU FINISH, CHECK EACH ONE

- ☐ I named two phases for the largest range and two for the smallest.
- ☐ I can explain why there are two bulges and not one.
- ☐ I wrote all three word meanings in my own words.

LEARN THE IDEA

23 Reading tidal data

Name _____

Date _____

BY THE END OF THIS LESSON I CAN

1. Calculate a tidal range by subtracting, and complete a data column.
2. Find a pattern in real data and use it to predict something I have not seen.

WORDS USED IN THIS LESSON

subtract to take one number away from another

pattern something that repeats in the data and can be pointed to

predict to say what will happen next, using a pattern

coincidence two things happening together by chance, with no real connection

READ Finding a pattern in real numbers

Read it all once. Then read it again and underline one sentence in each part.

Part 1

Patterns come from records

Scientists rarely spot a pattern by staring at the sea. They spot it by writing numbers down over time and then looking for what repeats.

The table on the next page records tide heights at one harbor on five dates, together with the moon phase on each date. To find the tidal range, you subtract the low tide height from the high tide height.

Part 2

How to hunt for the pattern

Once all the ranges are worked out, find the two largest and check what those dates have in common. Then look at the smallest and do the same.

Do the arithmetic for every row before you start looking. If you try to spot the pattern halfway through, you will usually find the one you were expecting rather than the one that is really there.

Part 3

The test of a real pattern

If a pattern is real, it should let you predict something you have not seen yet. That is how scientists tell a real pattern from a coincidence.

The tide tables printed for sailors are built on exactly this: a pattern found in old data, then used to predict the tides years ahead. The predictions are checked against what actually happens, which keeps the tables honest.

CHECK YOUR READING Every answer is in the passage. The part number tells you where to look.

1. How do you calculate a tidal range? (Part 1)

2. Why should you do all the subtractions before you look for the pattern? (Part 2)

3. What is the test of whether a pattern is real? (Part 3)

4. What are tide tables for sailors built from? (Part 3)

USE THE IDEA

23 Reading tidal data

WORKED EXAMPLE Study this before you write anything. Copy the method, not the answer.

Take October 1. High tide is 2.8 m and low tide is 0.3 m.

2.8 minus 0.3 equals 2.5, so the tidal range on October 1 is **2.5 m**. That row is filled in for you.

Do the same subtraction for every other row. Write each answer in meters, to one decimal place, exactly as the first row does.

NOW YOU TRY

STEP 1 Fill in the tidal range for the four empty rows.

Date	Moon phase	High tide (m)	Low tide (m)	Tidal range (m)
October 1	new moon	2.8	0.3	**2.5**
October 8	first quarter	2.1	0.9	
October 15	full moon	2.9	0.2	
October 22	last quarter	2.0	1.0	
October 29	new moon	2.7	0.3	

STEP 2 Which two dates have the largest range?

STEP 3 What do the moon phases on those two dates have in common?

STEP 4 Predict the tidal range at the next full moon, and explain your prediction.

IF YOU GET STUCK

Work out all five ranges first and list them down the side of the page. The pattern is much easier to see in five numbers than in a table of twenty-five.

CHALLENGE

The two new moon ranges are close but not identical. Does that break the pattern? Explain what you think and why.

BEFORE YOU FINISH, CHECK EACH ONE

- [ ] I filled in all four empty range boxes.
- [ ] My largest ranges are on the new moon and full moon dates.
- [ ] My prediction in Step 4 has a reason attached to it.

LEARN THE IDEA

24 How it all connects

Name _____

Date _____

BY THE END OF THIS LESSON I CAN

1. Build a concept map showing what causes what in the Earth, moon and Sun system.
2. Write sentences that name a cause and an effect.

WORDS USED IN THIS LESSON

system
a set of parts that work together and affect each other

concept map
boxes of ideas joined by labeled arrows

cause
the thing that makes something else happen

effect
the thing that happens as a result

READ One system, many effects

Read it all once. Then read it again and underline one sentence in each part.

Part 1

Four effects, one system

Everything in this unit comes from the same small set of motions. That is what makes it a system rather than a list of separate facts.

Earth spins on its axis, and that gives us day and night. Earth travels around the Sun, and the tilt combined with that travel gives us the seasons. The moon travels around Earth, and its changing position gives us the phases. The moon and the Sun pull on the oceans, and their changing alignment gives us the pattern of tidal ranges.

Part 2

What a concept map is

A concept map shows a system on paper. You write the important ideas in boxes, draw arrows between them, and on each arrow you write how one thing causes the other.

The words on the arrows are the whole point. A map with no words on its arrows shows that things are connected but not how, which is only half an explanation.

Part 3

Turning a map into a sentence

Once your map is drawn, every labeled arrow turns straight into a sentence. Read the box at the start of the arrow, then the words on the arrow, then the box at the end.

That is a useful check on your own work. If an arrow will not turn into a sentence that makes sense, the arrow is wrong or its label is missing.

CHECK YOUR READING Every answer is in the passage. The part number tells you where to look.

1. Which motion causes the seasons, and what else has to be true? (Part 1)

2. Name the four effects that come out of this one system. (Part 1)

3. Why is it not enough to draw arrows without writing on them? (Part 2)

4. How can you check whether an arrow on your map is right? (Part 3)

USE THE IDEA

24 How it all connects

WORKED EXAMPLE Study this before you write anything. Copy the method, not the answer.

One finished arrow: ROTATION → DAY AND NIGHT, with the words *spinning carries each place into and out of the sunlight* written on the arrow.

Turned into a sentence: Earth's rotation causes day and night, because spinning carries each place into and out of the sunlight.

Every arrow you draw should turn into a sentence like that one. If it will not, fix the arrow.

NOW YOU TRY

STEP 1 Draw an arrow from each cause to the effect it produces. Write on every arrow how it happens. Hint: two causes point to the same effect.

CAUSES	EFFECTS
ROTATION	DAY AND NIGHT
REVOLUTION	SEASONS
TILT OF THE AXIS	MOON PHASES
THE MOON'S ORBIT	TIDAL RANGE
PULL OF THE SUN AND MOON	

STEP 2 Write two sentences from your map. Each one must name a cause and an effect.

IF YOU GET STUCK

You need five arrows, and Part 1 of the reading gives you all of them. Read that paragraph one sentence at a time and draw one arrow for each cause you meet.

I CAN...

explain day and night · explain the seasons · name and place the four phases · explain why tidal range changes through the month

BEFORE YOU FINISH, CHECK EACH ONE

- [ ] Every cause box has an arrow coming out of it.
- [ ] Every arrow on my map has words written on it.
- [ ] Both of my sentences say what causes what.

LEARN THE IDEA

25 Weekly weather log

Name _____

Date _____

BY THE END OF THIS LESSON I CAN

1. Record the weather the same way every week for nine weeks.
2. Explain why one measurement can never show a pattern.

WORDS USED IN THIS LESSON

record — to write a measurement down at the time you take it

precipitation — rain, snow, sleet or hail falling from the sky

method — the way you take a measurement, which must stay the same

pattern — something that repeats across many measurements

READ Why one reading is never enough

Read it all once. Then read it again and underline one sentence in each part.

Part 1

One measurement tells you almost nothing

If you measure the temperature once, you know the temperature once. You know nothing at all about patterns.

A pattern only appears when you have measurements taken the same way, over and over, across a stretch of time. That is why scientists keep records for months, for years, and sometimes for more than a century. Some weather records used today were started by people who never saw what their data would be used for.

Part 2

Measuring the same way matters

Your job is to record the weather once a week, every week, using the same measurements each time.

Measuring the same way matters as much as measuring often. If you change your method halfway through, a difference in your data might come from the change of method rather than from the weather, and you would have no way of telling which.

Part 3

Keep going even when it feels pointless

Some weeks this will feel pointless, especially when the weather has barely changed. Keep doing it anyway.

By Unit 3 you will have a column of data collected by you, and you will be able to see something in it that no single day could ever have told you. That is the whole reason scientists keep records.

CHECK YOUR READING Every answer is in the passage. The part number tells you where to look.

1. Why is a single weather measurement not enough to find a pattern? (Part 1)

2. Why must you measure the same things in the same way each week? (Part 2)

3. How long do scientists sometimes keep weather records for? (Part 1)

4. What will you be able to do with your data by Unit 3? (Part 3)

USE THE IDEA

25 Weekly weather log

WORKED EXAMPLE Study this before you write anything. Copy the method, not the answer.

Here is one week filled in properly.

Week 1 · Monday · Temperature: 75°F · Sky: mostly cloudy · Wind: light breeze · Precipitation: none

Every box has something in it, including the ones where nothing happened. Writing *none* is data. An empty box is not, because later you cannot tell whether it means none or means you forgot.

NOW YOU TRY

STEP 1 Fill in one row each week, on the same day of the week, at about the same time of day.

Week	Date	Temperature	Sky / clouds	Wind	Precipitation
1					
2					
3					
4					
5					
6					
7					
8					
9					

STEP 2 At the end of nine weeks: what pattern do you notice in your data?

IF YOU GET STUCK

No thermometer? Describe the temperature instead: cold enough for a coat, warm enough for short sleeves. Use the same descriptions every week and the log still works.

CHALLENGE

Compare your log with a partner's. Where your entries differ for the same week, work out whether the weather really differed or whether you measured it differently.

BEFORE YOU FINISH, CHECK EACH ONE

- ☐ I filled in every box in my row, writing none where nothing happened.
- ☐ I measured at about the same time of day each week.
- ☐ I used the same words each week to describe the sky and the wind.

UNIT REVIEW · LESSONS 15 TO 25

Show what you know

Name _____

Date _____

PART A · VOCABULARY. WRITE THE LETTER OF THE MATCHING MEANING.

____ 1. rotation	A. how much of the moon's lit half we can see
____ 2. revolution	B. Earth spinning on its axis once a day
____ 3. tilt	C. light spread thin over a large area
____ 4. phase	D. Earth's yearly trip around the Sun
____ 5. tidal range	E. high tide height minus low tide height
____ 6. indirect rays	F. the lean of Earth's axis

PART B · CHOOSE THE BEST ANSWER. CIRCLE THE LETTER.

7. We see a full moon when the moon is
   A. between Earth and the Sun B. on the far side of Earth from the Sun C. inside Earth's shadow
8. It is summer in Virginia. In Australia it is
   A. summer B. winter C. spring D. the same season
9. The largest tidal ranges happen at
   A. new moon and full moon B. the two quarter phases C. every high tide
10. Which list is in the correct order?
    A. new, full, first quarter, last quarter B. new, first quarter, full, last quarter C. full, new, last quarter, first quarter

PART C · USE THE DATA.

At one harbor, high tide was 3.1 m and low tide was 0.4 m.

11. What was the tidal range? Show your subtraction.

_____

12. That day had one of the largest ranges of the month. Name the two phases the moon could have been in, and give one reason.

_____

_____

UNIT REVIEW

Explain like a scientist

PART D · FIX THE DIAGRAM

13. In the Lesson 20 diagram, a student shaded the moon at position B so that its lit half faces **up, toward Earth**. Is the drawing right or wrong? Explain using the rule from Lesson 20.

PART E · CLAIM, EVIDENCE, REASONING

14. Your cousin says: *"We have summer because Earth gets closer to the Sun in June."* Write a scientific response. Use all three parts.

CLAIM

What is the real cause of the seasons? Answer the question in one sentence.

The seasons are caused by _____

EVIDENCE

What fact proves distance is not the cause? Use something you learned in Lesson 17.

One piece of evidence is that _____

REASONING

Explain how your evidence supports your claim. Use the words tilt, direct and revolution.

This shows that _____

	3 points	2 points	1 point
Claim	Names tilt and revolution together	Names only one of the two	Names distance or no cause
Evidence	Uses opposite seasons in the two hemispheres, stated accurately	Gives a true fact that only partly helps	Evidence is missing or inaccurate
Reasoning	Connects the evidence to the claim using tilt, direct rays and revolution	Connects them but misses a key word	Restates the claim without explaining

BEFORE THE TEST, CHECK EACH ONE

- ☐ I checked Parts A to D against the answer key and fixed every mistake with a note.
- ☐ I scored my CER response with the rubric and rewrote the weakest part.
- ☐ I reread the Careful boxes in Lessons 15, 17, 20 and 22.

Glossary

Every word taught in this unit, with the lesson where you met it first.

alignment L22
how the Sun, Earth and moon are lined up

away L17
leaning in the opposite direction from something

bulge L22
a place where ocean water is pulled up higher

cause L24
the thing that makes something else happen

coincidence L23
two things happening together by chance, with no real connection

concentrated L18
packed into a small space

concept map L24
boxes of ideas joined by labeled arrows

correct L21
to find a mistake and write what it should have been

cycle L19
a set of changes that repeats in the same order

direct rays L18
light arriving straight down, packed onto a small area

effect L24
the thing that happens as a result

full moon L19
the phase when we see all of the lit half

hemisphere L17
one half of Earth, the northern half or the southern half

indirect rays L18
light arriving at a slant, spread thin over a large area

lit half L20
the half of the moon the Sun shines on

lunar eclipse L21
the rare event when Earth's shadow falls on the moon

method L25
the way you take a measurement, which must stay the same

midday L16
when your place faces the Sun most directly

midnight L16
when your place is turned farthest from the sunlight

new moon L19
the phase when we cannot see the lit half at all

orbit L20
the path of one object around another

pattern L23, 25
something that repeats in data and can be pointed to

phase L19
how much of the moon's lit half we can see from Earth

position L16, 20
where something is at one particular moment

precipitation L25
rain, snow, sleet or hail falling from the sky

predict L23
to say what will happen next, using a pattern

record L25
to write a measurement down at the time you take it

revolution L17
Earth's yearly trip around the Sun

rotation L15
Earth spinning on its axis, once in about 24 hours

shadow L21
the dark area behind something that is blocking light

spread L18
shared out over a larger space, so each part gets less

subtract L23
to take one number away from another

sunrise L15
the moment your place turns into the sunlight

sunset L15
the moment your place turns out of the sunlight

system L24
a set of parts that work together and affect each other

tidal range L22
the difference in height between high and low tide

tide L22
the daily rise and fall of the sea along a coast

tilt L17
the lean of Earth's axis, which points the same way all year

time zone L15, 16
an area of Earth that keeps the same clock time

toward L17
leaning in the direction of something

viewing angle L20, 21
the direction we look at something from

Answer key

Read this before you use these pages. Do the lesson first. Then check. Then fix anything wrong in a different color and write a short note saying why it was wrong. If your wording is different but your idea matches, you are right. Answers marked *your own answer* have no single right response.

Lesson 15 · Day and night

Check. 1. Earth. It is rotating; the Sun is not moving across our sky. 2. Half of it. 3. Your place is turning out of the sunlight. 4. Different places are at different points in the same rotation, so they reach the sunlight at different moments.

Step 1. Left half DAY, right half NIGHT. The dashed line is the axis; the curved arrow shows the direction of rotation.

Step 2. We have day and night because Earth rotates on its axis, carrying each place into and out of the sunlight.

Step 3. No. Places elsewhere face the Sun at different moments of the rotation, so their clocks show different times.

Lesson 16 · Where am I when it is night?

Check. 1. When it faces the lamp most directly. 2. It is turning into the light. 3. Each place reaches the light at a different moment; without time zones, midday would fall in the middle of the night somewhere. 4. No. The globe turning is enough on its own.

Table. Position 2: lit edge, about sunset, the dot is leaving the lit half. Position 3: dark, about midnight, the dot is turned farthest from the sunlight. Position 4: dark edge, about sunrise, the dot is about to enter the lit half.

Challenge. East. Places to the east turn into the sunlight earlier, so their day is further along.

Lesson 17 · Why do we have seasons?

Check. 1. Winter. 2. The whole planet is about the same distance from the Sun at any moment, so if distance were the cause, the whole planet would have one season at once. It does not. 3. Earth's tilt together with its revolution. 4. No, the tilt points the same way all year. What changes is where Earth is on its orbit, and so which half leans toward the Sun.

Step 1. A Northern summer (top of axis leans toward the Sun). B Fall. C Northern winter (leans away). D Spring.

Step 2. Winter. In June the southern half, where Australia is, leans away from the Sun, so it gets indirect rays and short days.

Lesson 18 · Direct and indirect rays

Check. 1. The light does not get weaker; the same light is spread over a larger area, so each part gets less. 2. Direct rays, because the light is concentrated on a smaller area. 3. No. Only the angle at which sunlight arrives changes. 4. Longer days: more time to warm up and less time to cool at night.

Table. A: 4 squares, dark gray, A's squares get more light each, A heats more, A is like summer. B: 8 squares, light gray.

Step 2. DIRECT rays and LONGER days.

Challenge. At midday the Sun is highest, so its rays are more direct. At breakfast the Sun is low and its rays are indirect, spread over more ground.

Spiral Check 1

1. B, rotation (Lesson 15).

2. No. Japan is at a different point in the same rotation, so it is facing the Sun at a different moment (Lessons 15 and 16).

3. When it is summer in Virginia it is winter in Australia, and both are the same distance from the Sun (Lesson 17).

4. A and C, tilt and revolution (Lesson 17).

5. …spread over a larger area by indirect rays, so each part gets less (Lesson 18).

6. About midnight, because midnight is when a place is turned farthest from the sunlight (Lesson 16).

Lesson 19 · The four moon phases

Check. 1. No. It is always a ball with half of it lit. What changes is how much of that lit half we can see. 2. New moon, first quarter, full moon, last quarter. 3. About 29 days, about a month. 4. We always see the same side, so the far side never faces us.

Steps 1 and 2. 1 new moon, 2 first quarter, 3 full moon, 4 last quarter, left to right.

Step 3. Your own words, matching the meanings in the word list.

Lesson 20 · Moon positions and phases

Check. 1. It always faces the Sun. That never changes. 2. Between Earth and the Sun. 3. On the far side of Earth from the Sun. 4. Shading the lit half toward Earth instead of toward the Sun.

Table. B first quarter: the lit half faces the Sun, and from Earth we see it from the side, half lit. C full moon: opposite the Sun, so the whole lit half faces Earth. D last quarter: we see the other lit half from the side.

Lesson 21 · Fix the diagram

Check. 1. The moon would look the same every night, because Earth's shadow always points away from the Sun. 2. A lunar eclipse; a few times a year at most, lasting a few hours. 3. Our changing viewing angle as the moon travels around Earth. 4. Take the idea, follow it to what we would see if it were true, and compare that with what we actually see.

Steps 1 and 2. Position 2 should be first quarter, not full moon. Position 3 should be full moon, not first quarter.

Step 3. Earth's shadow always points the same way, so it could not make a moon that changes through the month.

Challenge. About 14 or 15 days: new to full is half the cycle, and half of 29 is about 14 and a half.

Lesson 22 · Tides

Check. 1. There are two bulges, one facing the moon and one on the opposite side; as Earth rotates, each coast passes through both. 2. The difference between high and low tide heights; subtract low from high. 3. The Sun, Earth and moon are lined up, so the two pulls work together. 4. First quarter and last quarter.

Steps 1 and 2. Largest: new moon and full moon. Smallest: first quarter and last quarter.

Lesson 23 · Reading tidal data

Check. 1. Subtract the low tide height from the high tide height. 2. Looking halfway through makes you find the pattern you expected instead of the real one. 3. A real pattern lets you predict something you have not seen yet. 4. A pattern found in old data, used to predict tides years ahead.

Step 1. October 8: 1.2 m. October 15: 2.7 m. October 22: 1.0 m. October 29: 2.4 m.

Steps 2 and 3. October 15 and October 1. Full moon and new moon: the Sun, Earth and moon are lined up.

Step 4. About 2.5 to 2.8 m, because full moons in this data give the largest ranges.

Challenge. No. Both new moon ranges are among the largest in the table. Real data varies a little; a pattern means a repeated trend, not identical numbers.

Lesson 24 · How it all connects

Check. 1. Revolution around the Sun, and Earth's axis must also be tilted. 2. Day and night, the seasons, the moon phases, the pattern of tidal ranges. 3. Arrows without words show that things are connected but not how. 4. Try to turn it into a sentence. If it will not make sense, the arrow is wrong or its label is missing.

Map. ROTATION to DAY AND NIGHT. REVOLUTION to SEASONS. TILT OF THE AXIS to SEASONS. THE MOON'S ORBIT to MOON PHASES. PULL OF THE SUN AND MOON to TIDAL RANGE. The words on your arrows are your own.

Lesson 25 · Weekly weather log

Check. 1. One measurement tells you the weather once; a pattern only appears across many measurements taken the same way. 2. A change of method could cause a difference in the data, and you could not tell it from a change in the weather. 3. Months, years, and sometimes more than a century. 4. See a pattern in your own data that no single day could show.

Log. Your own data. Check that no box is empty and that you wrote none where nothing happened.

Unit Review · Parts A to C

Part A. 1 B · 2 D · 3 F · 4 A · 5 E · 6 C

Part B. 7 B · 8 B · 9 A · 10 B

11. 3.1 minus 0.4 equals 2.7 m.

12. New moon or full moon, because the Sun, Earth and moon are lined up and their pulls work together.

Unit Review · Parts D and E

13. Wrong. The lit half always faces the Sun, so at position B it must face left, toward the Sun. Shading it toward Earth is the most common mistake named in Lesson 20.

14. Model response. *Claim:* The seasons are caused by Earth's tilt together with its revolution around the Sun, not by distance. *Evidence:* When it is summer in Virginia it is winter in Australia, even though both places are the same distance from the Sun. *Reasoning:* If distance caused the seasons, the whole planet would have the same season at once. Instead, as Earth revolves, the tilted half leaning toward the Sun gets direct rays and longer days, so it has summer while the other half has winter.

Scoring. Use the rubric on the review page. A full-credit answer names both tilt and revolution in the claim.

WHAT TO DO WITH A WRONG ANSWER

Cross it out, write the correct answer beside it, and add one short note beginning *I was wrong because…* Then reread the part of the lesson named in the question. Those notes are the best revision you can bring to the unit test.